Week 3 Problem Set: Limits at Infinity and Vertical Asymptotes

Calculus Honors · Module 2: Limits Without Bound & Special Limits · Lessons 2.01 and 2.02

Name: _____ Date: _____

> Show all of your work. A correct answer with no reasoning earns partial credit at most. Keep your writing neat and vertical, line up your equal signs, and label your final answers.

PART A · LIMITS AT INFINITY

1. For each limit, state which of the three degree-comparison cases applies, then evaluate it.

(a) $\lim_{x \to \infty} \dfrac{4x^3 - 2x}{2x^3 + 7}$ (b) $\lim_{x \to \infty} \dfrac{5x + 1}{3x^2 - 4}$ (c) $\lim_{x \to -\infty} \dfrac{2x^2 - 3}{x - 5}$

2. Evaluate $\lim_{x \to \infty} \dfrac{7x^2 + x - 1}{4x^2 - 9}$ by dividing every term by the highest power of x. Show every step of the division before you take the limit.

PART B · INFINITE LIMITS AND VERTICAL ASYMPTOTES

3. Let $f(x) = \dfrac{5}{x+2}$. (a) Find $\lim_{x \to -2^-} f(x)$ and $\lim_{x \to -2^+} f(x)$ using a sign test at a nearby point on each side. (b) Does $\lim_{x \to -2} f(x)$ exist? Explain, and state the vertical asymptote.

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4. Let $g(x) = \dfrac{-3}{(x-1)^2}$. Determine the one-sided and two-sided behavior of g as $x \to 1$, and state whether $x = 1$ is a vertical asymptote. Explain why the squared denominator matters here.

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5. Let $h(x) = \dfrac{x+5}{x^2-25}$. Simplify h(x) for $x \neq -5$, then evaluate $\lim_{x \to -5} h(x)$. Explain why the zero denominator at $x = -5$ does not create a vertical asymptote.

PART C · READING A GRAPH

6. The graph of $y = k(x)$ has a vertical asymptote at $x = 4$. Immediately to the left of $x = 4$ the curve rises without bound, and immediately to the right it also rises without bound. State $\lim_{x \to 4^-} k(x)$, $\lim_{x \to 4^+} k(x)$, and $\lim_{x \to 4} k(x)$ with correct notation. Does a vertical asymptote require the two one-sided limits to agree? Explain.

PART D · EXPLAIN YOUR THINKING

7. Convince a classmate why a function can have a vertical asymptote at $x = a$ even though, strictly speaking, "the limit does not exist there" in the sense of approaching a single finite number. Use one of the examples above as evidence.